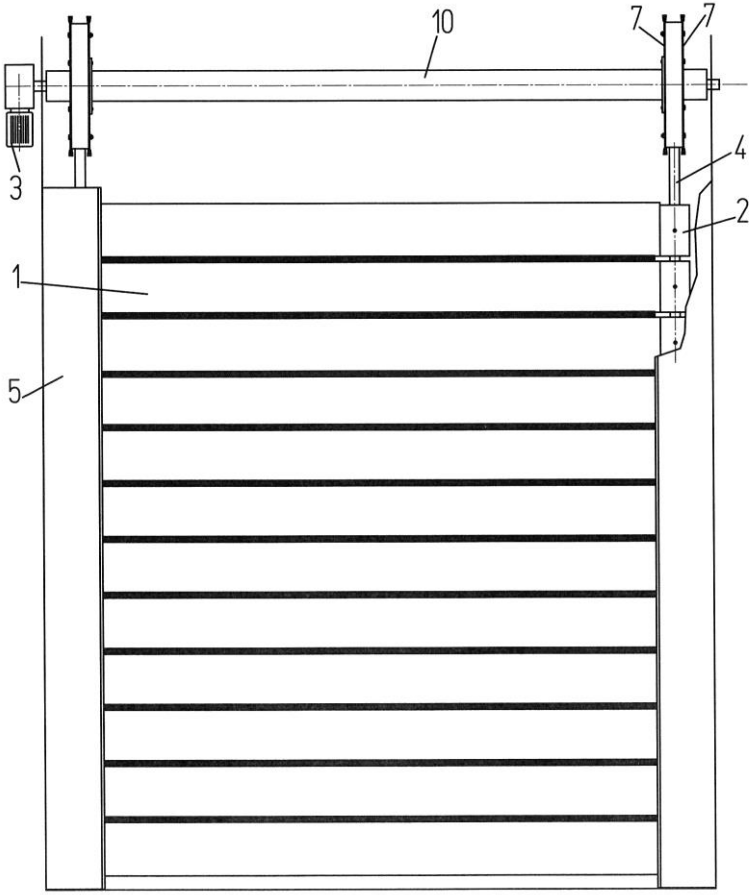
Albany RR3000 ISO

THE NEXT GENERATION OF HIGH SPEED DOORS



General design RR3000 ISO

- Side frame rh / lh (5) with guiding and spring box
- Top roll (10) with spiral disks (7)
- Door blade with slats (1), end pieces (2) and flat belt (4)
- Drive (3) and control system
- Balancing system with tension springs (in side frames)
- Safety devices (not shown)



Superior size

- Daylight width: up to 7m
- Daylight height: up to 6m
- max. size: 42 m²

Bigger than current design of RR3000 (max. 36 m^2)

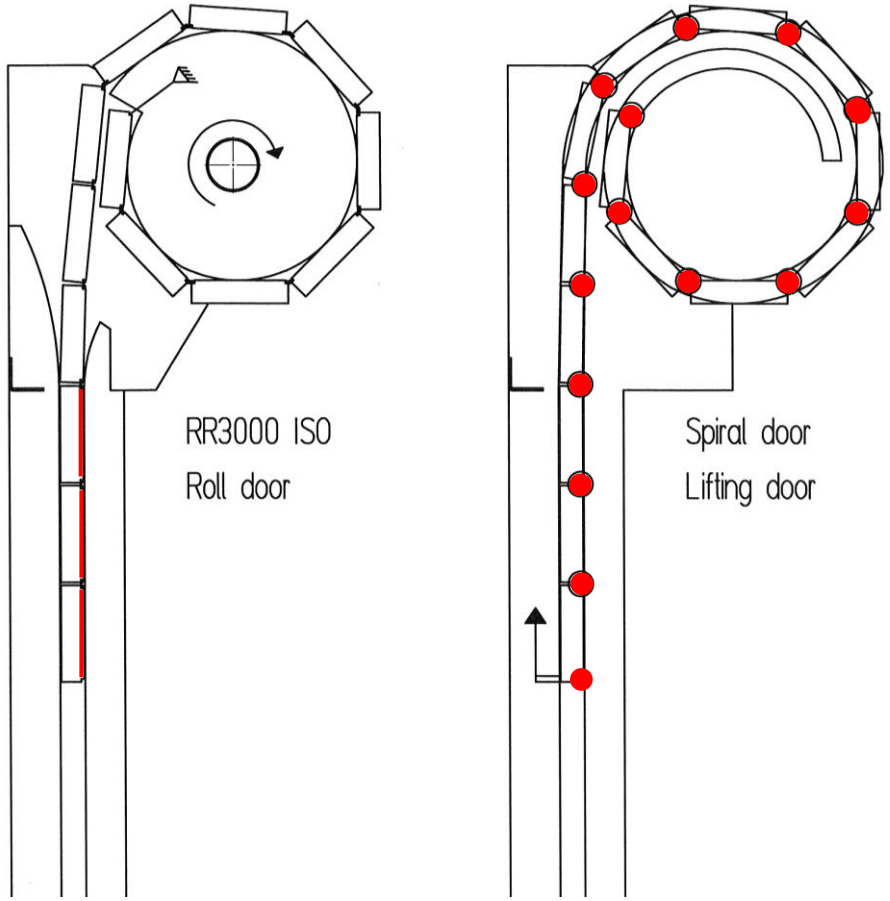
and most competitors (Hörmann: appr. 36m^2)



Roll door vs Spiral doors

Comparison - moving parts

- Spiral door: all lamellas and rollers are permanently active
- RR3000 ISO: no rollers, no relative movement on top roll -> 40% less operation



red: moving parts



Functionality

Superior control system

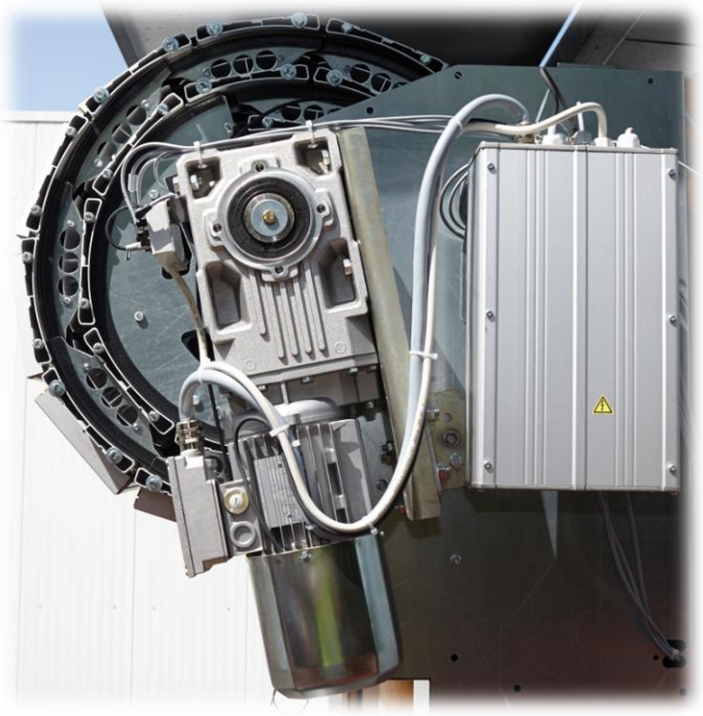
- **MCC** Vector control
- Proven reliability and performance
- Modular system
- High speeds (2,1/0,7 m/s)
- High operation cycles in continuous operation
- Smooth operation
- Optional equipment available:

-> Combi boxes (Plastic, Steel, Stainless)

-> UPS system

-> Bus systems

- Standard installation next to drive unit

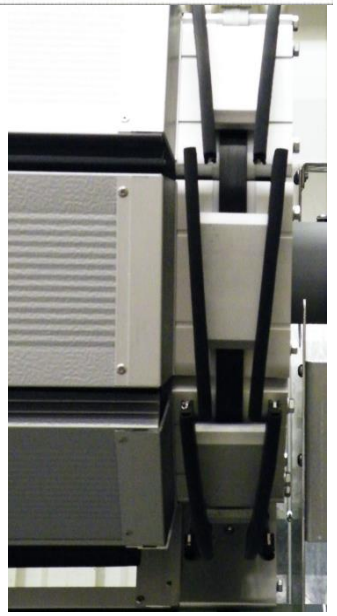


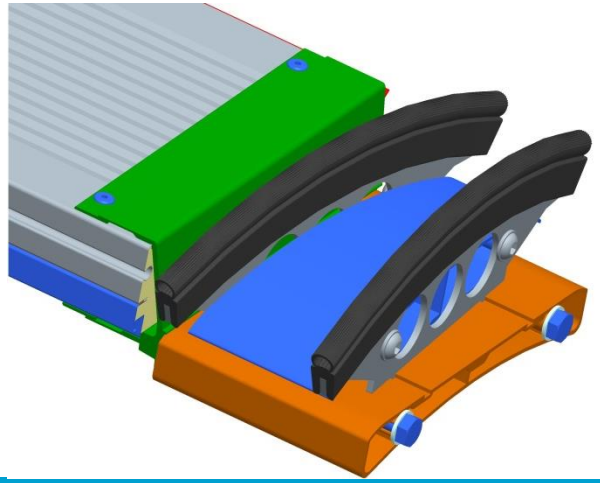
Patented "**V-Drive**" system

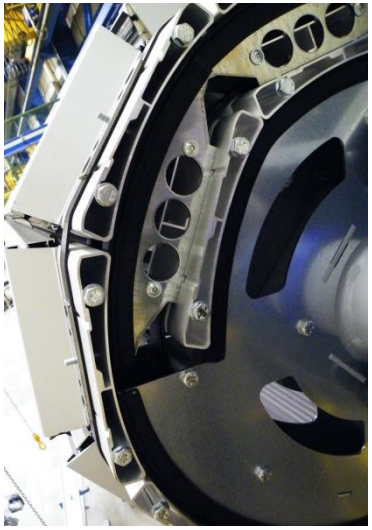
NIHVA

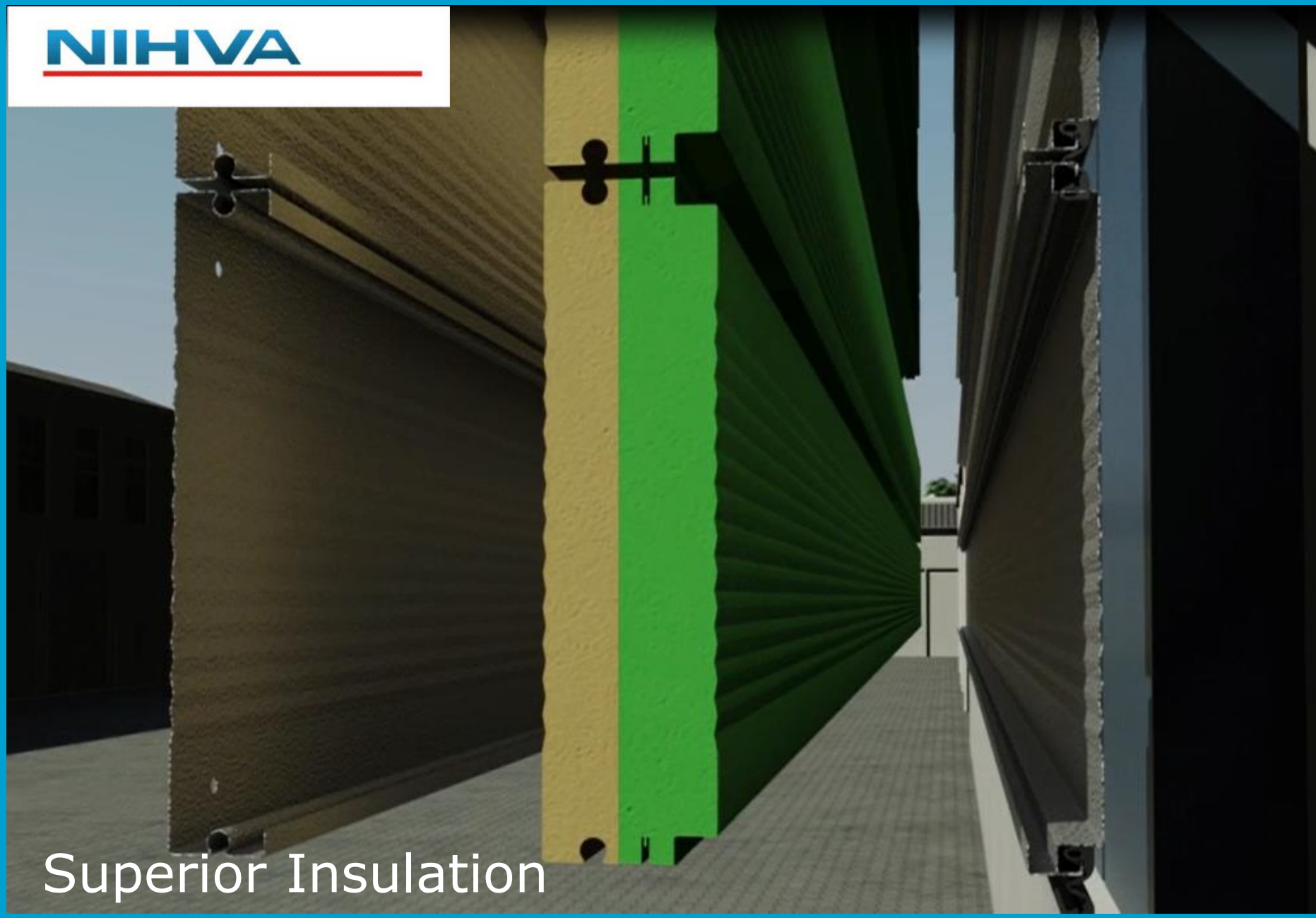
- Roll door
  - Door blade fixed at top roll (forces lead from top to bottom)
  - Minimum no of operating parts
- First layer of lamellas roll up on spiral disks
- Next lamellas roll up on special distance pieces (V-structure)

  -> Lamellas roll up touchless








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Superior Insulation

Superior insulation

- Insulated sandwich lamella with 50mm thickness, height 200mm
  (Hörmann: 42mm, Efaflex 40 at DW<=4500)
- Integrated thermal break (not at Hörmann)
- Dual gaskets between lamellas
- U-value lamella: 0.42 W/m²K
- U-value door blade: 1.4 W/m²K
- U value complete door incl. side frames:
  2.0 W/m²K **(6000x4700 mm without windows)***
- Sound reduction: 25dB



*= Specific U-value according to customer door configuration on request



Superior lamelle design

Superior lamella design

- Foam core covered by stable steel sheet covers
  - Minimized risk of dents
  - Minimized thermal deflection
    (Efaflex: Aluminum)
- High mechanical strength (stucco + grooves)
  - Wind class 2 up to DW 7m
    - Wind class 3 up to DW 5m
      (Hörmann, Efaflex at DW 5m: class 2)
  - Robust against scratches



- Standard color RAL 9006 (white aluminum)
  - Option 1: painting outside
  - Option 2: painting inside



Superior lamella design

Double glass window lamella

- Burglar resistant transparent plastic lamellas
- Aluminum anodized frame profiles
- Transparency: 65%
- 3 Windows = 5 Windows in previous model
- Window panes without web
- Quantity according customer wishes





Maximum window qty

top



bottom

Superior Endurance



High speed, high frequency, long lifetime

- Minimum operating parts
- No rollers, no hinges
- Long life tension springs
- Minimum maintainance requirement
- High level of pre-assembling ensures manufacturer quality and low installation costs
- Multi-million test cycles accomplished





Maintainance and spare parts

Major wear parts:

- Springs, straps, pulleys
- Damping profile

Recommended change intervals:

DH<=4000: 400.000 cycles
DH> 4000: 200.000 cycles



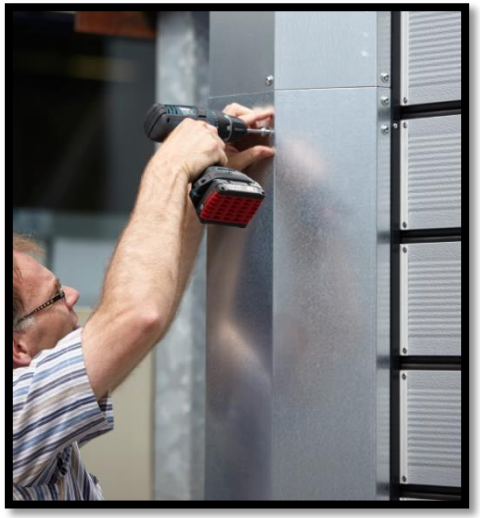


- Flat belts (right = left):

Lifetime: min. 500.000 cycles

Quick and easy repair

- Exchange of separate lamellas within 15 minutes
- Each lamella has same length (Easy stock keeping)





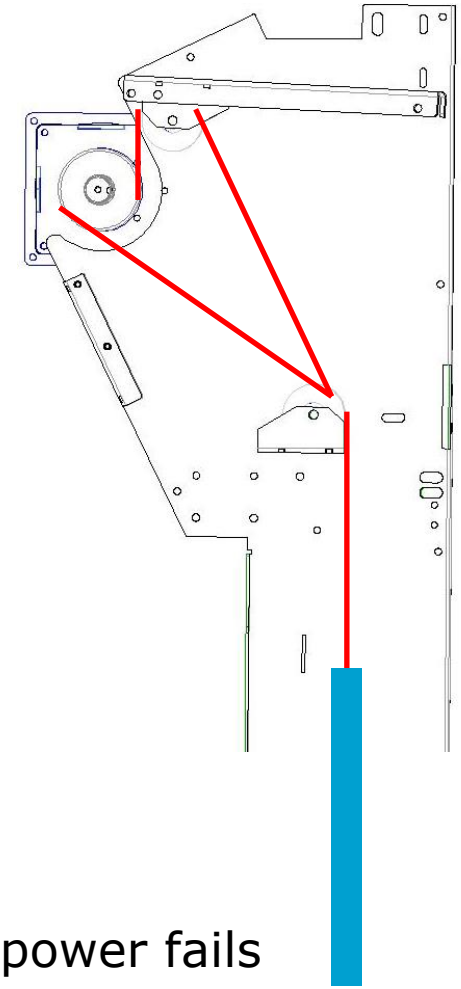






Superior balancing system





- Supporting drive unit
- Providing manual opening by lever when power fails

Superior cover design

- Material aluminum anodized -> low weight, weather resistant
- Additional support profiles stabilizes the covers
- Inspection openings allow easy access to components
- Two sizes depending on door dimensions -> reduces required space over lintel







Superior cover design - dimensions



DH	V
<=4700	920
>4700	1090

Safety devices

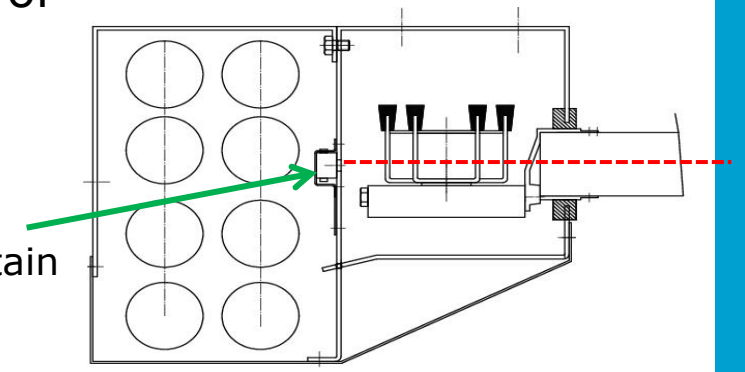


Safety devices

- Light grid (2.5 m) integrated in side frame
- Door fully balanced
- Double lifting belt at top lamellas
- Options:
  - Electrical safety edge + stationary photocell
  - LED traffic light strip
  - Hand chain operator
  - Huge bandwidth of external safety devices
  - Crash sensor: detects deformation of panel and stops door



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Crash-Sensor (optional)

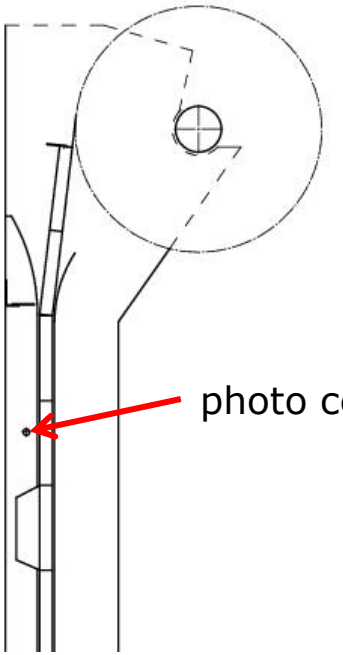
Detection of deformation on door blade outwards (e.g. after crash)

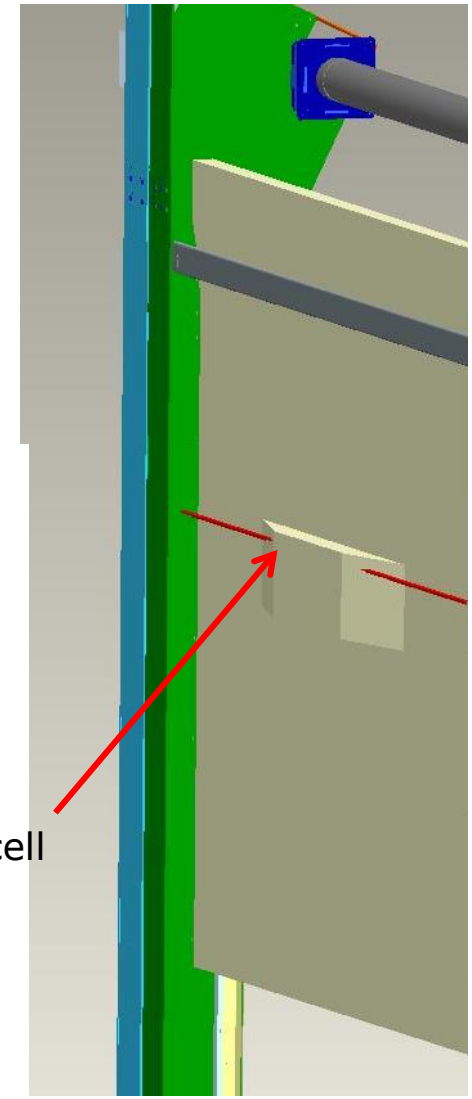
-> MCC stops opening

-> Only closing is possible

-> No crash at lintel profile







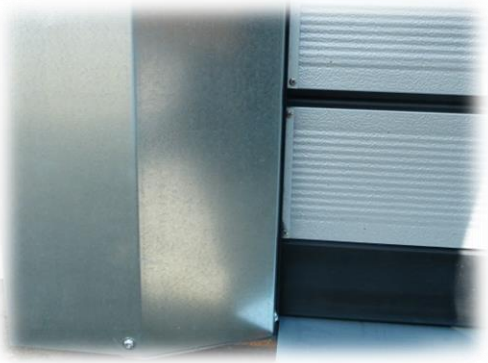


Improvements

Improvements vs. previous design

- Integrated light curtain
- Electrical safety edge (option)
  - No reduction of daylight opening
  - Same speed and rubber profile
  - Retrofit possible
- Simplified side frame design
- V-drive (modular concept, allowing bigger sizes)
- Improved sealing (dual sealing on panel, top lamella)
- Improved sealing (bottom sealing)
  - Bag instead lip
  - Finger protection
- merged with proven concept of RR3000 family





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Impressions



Customer: Warsteiner (7000x5000 mm)



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Test door (7000x6000 mm)



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